

Introduction to Machine Learning (SS 2024)

Programming Project

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I. INTRODUCTION

Detecting fraudulent transactions is a critical challenge in the financial industry, where even a small proportion of undetected fraud can result in substantial monetary losses. In this project, we formulate the problem as a binary classification task: predicting whether each transaction is legitimate (Class 0) or fraudulent (Class 1).

The dataset contains 45,569 records with 29 attributes, including a time feature ("Time"), 28 anonymized numerical features ("Feature0"–"Feature27"), the transaction amount ("Amount"), and the target variable ("Class"). Due to the extreme class imbalance (fraud rate $\approx 0.17\%$), overall accuracy is not a reliable performance measure. Instead, we will evaluate models using precision, recall, and area under the ROC curve (AUC). All models are trained in a supervised learning framework and assessed on a validation set that preserves the original class distribution, ensuring realistic estimates of generalization performance.

II. IMPLEMENTATION / ML PROCESS

Data Loading & Preprocessing. We load the transactions data from a compressed CSV and immediately drop the "Time" column, as it conveys only sequence order without predictive value. The remaining 28 anonymized features and the "Amount" column are separated from the target variable "Class" (0 = legitimate, 1 = fraudulent). Since all anonymized features are already normalized, we apply a `StandardScaler` only to "Amount" to harmonize its scale. A stratified train-validation split (80%/20%) preserves the original fraud ratio ($\approx 0.17\%$) in both subsets, ensuring reliable performance estimates.

Model Selection. We selected two classifiers with complementary strengths to capture both linear and non-linear patterns in the data. Logistic Regression offers a transparent baseline whose learned coefficients can be directly interpreted in terms of log-odds, supporting explainability in a high-stakes domain. Random Forest, by contrast, constructs an ensemble of decision trees to model complex feature interactions and mitigate overfitting through bootstrap aggregation.

Hyperparameter Optimization. To refine each model's predictive power, we conducted extensive empirical tuning of parameter ranges through repeated training:

- **Logistic Regression:**

Grid search over

- $C \in [10^{-3.5}, 10^{-1.5}]$
- `class_weight` $\in \{\text{None}, \text{"balanced"}\}$

using 5-fold cross-validation with ROC-AUC as the scoring metric. The bounds of the C grid were determined experimentally by observing convergence behavior and generalization performance across multiple pilot runs.

- **Random Forest:**

Randomized search sampling 20 configurations of

- `n_estimators` $\in \{200, 250, 300\}$
- `max_depth` $\in \{5, 8, 10\}$
- `min_samples_split` $\in \{3, 4, 5, 6\}$
- `min_samples_leaf` $\in \{2, 3, 4\}$
- `max_features` $\in \{\text{sqrt}, \text{log2}, \text{None}\}$
- `class_weight` set to "balanced"

using 4-fold cross-validation and ROC-AUC. Parameter ranges were iteratively narrowed based on performance patterns observed in preliminary experiments to balance computational cost and model expressiveness.

III. RESULTS

Our best Logistic Regression model (regularization parameter $C = 0.00292$, `class_weight=None`) and Random Forest model (`n_estimators = 300`, `max_depth = 8`, `min_samples_split = 5`, `min_samples_leaf = 3`, `max_features = "log2"`, `class_weight="balanced"`) both achieved a ROC-AUC of 0.984 on the hidden test set.

On our independent 20/80 stratified split, Logistic Regression achieved a ROC-AUC of 0.9840, while Random Forest reached 0.9844. In the original Jupyter notebook evaluation, Logistic Regression scored 0.9830 on the training set and 0.9877 on the hidden test set; Random Forest scored 0.9968 on the training set and 0.9828 on the hidden test set. These consistently high AUC values across both evaluation protocols confirm the robustness and generalization capability of both classifiers.

Precision–Recall analysis (Fig. 1) confirms the superior average precision (AP) of Random Forest (AP = 0.873) over Logistic Regression (AP = 0.796), demonstrating more robust fraud detection at high-recall operating points. The ROC curves for both models are virtually identical (Fig. 2), and both significantly outperform random guessing.

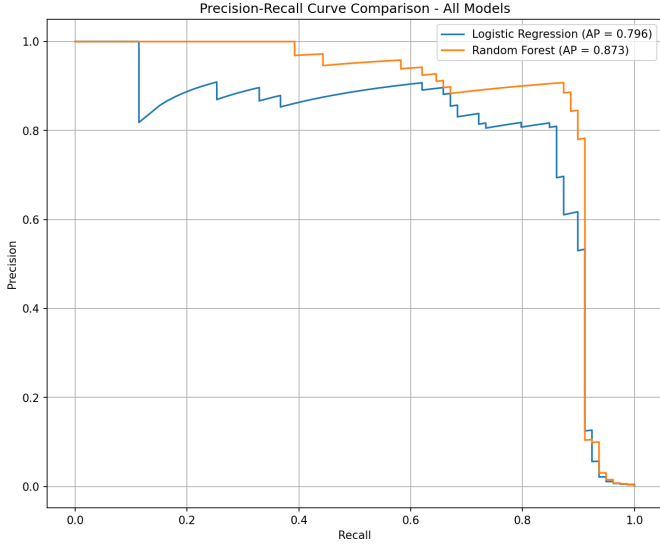


Fig. 1: Precision–Recall Curve Comparison

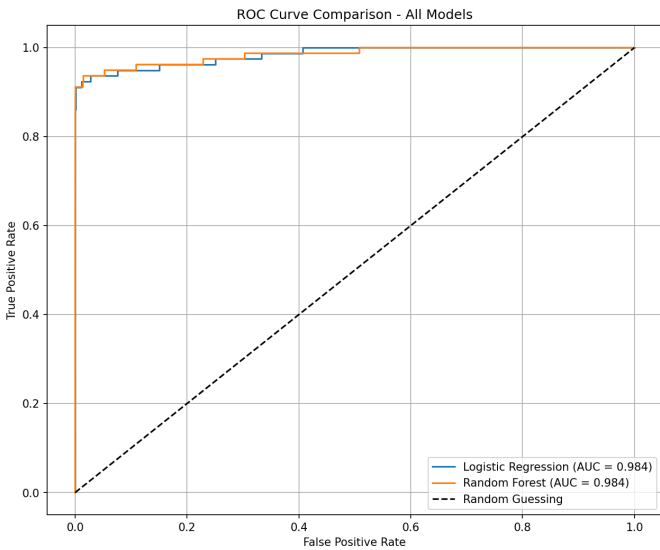


Fig. 2: ROC Curve Comparison

Confusion matrices plotted at each model’s optimal decision threshold (Fig. 3) show that Random Forest correctly identifies 69 of 79 fraud cases with 10 false negatives, while Logistic Regression correctly identifies 67 with 12 false negatives. Both models incur fewer than 20 false positives out of over 45 000 legitimate transactions, illustrating a favorable trade-off between precision and recall.

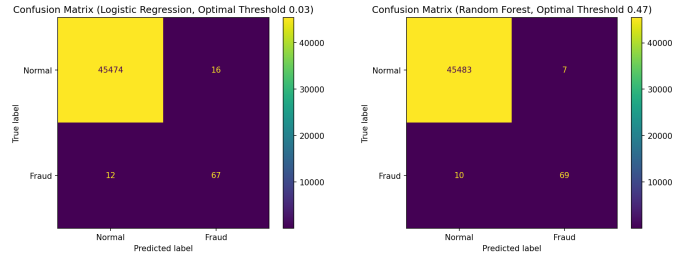


Fig. 3: Confusion Matrices at Optimal Thresholds: (a) Logistic Regression, (b) Random Forest.

IV. DISCUSSION

The evaluation confirms that both Logistic Regression and Random Forest yield excellent discrimination between legitimate and fraudulent transactions, with ROC–AUC values consistently around 0.984 across independent splits and unseen test sets. While Random Forest demonstrated a higher average precision (0.873 vs. 0.796), indicating stronger performance at higher recall thresholds, Logistic Regression exhibited notably stable behavior, achieving nearly identical AUCs with far fewer model parameters and training complexity.

The confusion matrices reveal that Random Forest correctly captures 69 out of 79 fraud cases with 10 false negatives, whereas Logistic Regression identifies 67 frauds with 12 false negatives. Both incur fewer than 20 false positives among over 45000 legitimate transactions, suggesting a precision–recall trade-off well suited for risk-averse settings. In operational terms, Random Forest’s slightly higher recall may be preferred when missing a fraudulent transaction carries a high cost, whereas Logistic Regression’s simplicity and interpretability could facilitate regulatory compliance and faster decision-making.

The very low fraud rate ($\approx 0.17\%$) makes choosing an appropriate decision threshold essential; future work could explore cost-sensitive algorithms or targeted oversampling to improve detection. Because the features are anonymized, there is limited opportunity for domain-driven feature engineering—access to raw transaction details or time-series patterns might further boost performance. Finally, more complex models such as gradient-boosted trees or neural networks may yield additional gains, though at the expense of increased training time and interpretability.

V. CONCLUSION

Both Logistic Regression and Random Forest demonstrate robust fraud detection, each achieving ROC–AUC 0.984 and fewer than 20 false positives per 45 000 legitimate transactions. Logistic Regression offers transparency and lower computational cost, while Random Forest provides a slight edge in recall. The optimal choice depends on the organization’s priorities for interpretability, processing resources, and tolerance for missed fraud.